

EXPERIENCE

Bloomberg LP — Fleet Automation Services Platform | Senior Software Engineer June 2025 – Present

- Led expansion of Bloomberg's fleet automation platform to 6,000 Juniper routers, replacing manual SSH upgrades requiring up to 3,000 engineer-days per quarterly cycle.
- Co-designed a Go-based orchestration platform spanning 7 services and 100,000+ machines, evolving batch workflows towards Kafka-driven events to reduce stale state and race conditions across eligibility, scheduling, ticketing, and reconciliation.
- Designed and implemented a DORA-driven security campaign across 60,000 machines and 6 infrastructure teams, extending fleet orchestration with multi-stage rollouts, telemetry prechecks, and campaign-specific scheduling constraints.
- Led 2 fleet automation campaigns end-to-end, translating infrastructure requirements into technical proposals, architecture, implementation, and production support.
- Led a 4-engineer hackathon team to build a Bloomberg Philanthropy MCP server, integrating IAM-protected services to surface events and volunteer activity through an LLM interface; productionized for internal use.

Bloomberg LP — ESG Analytics & Government Climate Risk | Software Engineer Jan 2024 – June 2025

- Rebuilt Bloomberg's Government Climate Risk scoring pipeline for 151 countries, replacing 24+ runbook-driven Python scripts with an Argo-orchestrated service and reducing score computation from 2 weeks of engineer effort to 15 minutes.
- Established a regression-tested migration path for historical climate risk scores back to 2015, reproducing legacy results while transitioning PostgreSQL-to-pandas transformation workflows to the new pipeline.
- Introduced Pandera-based validation for semantically invalid model inputs and established dataframe testing patterns adopted across other pandas-based ESG teams.
- Automated score publication to Bloomberg Terminal and data licensing consumers, reducing a day-long legacy workflow to <10 minutes and enabling push-button publication.
- Independently built and deployed a file diff tool used by ~100 internal quants, analysts, product managers, and engineers to identify erroneous tabular and nested JSON model inputs before production scoring.

Bloomberg LP — ESG Score Computation | Software Engineer May 2022 – Jan 2024

- Replaced ad hoc Python scripts with 6 microservices, scaling ESG scoring from monthly manual processing of 3,000 companies to automated nightly scoring of 15,000.
- Built a snapshot service versioning 1M+ nightly data points, enabling exact point-in-time score reproduction while persisting only ~5% of changed values through time-series diffing.
- Increased scoring throughput by 80x, processing 15,000 companies in under 1 hour vs. 3,000 in 16 hours by decomposing 4 scoring models, normalizing PostgreSQL schemas and optimizing indexes and queries.
- Simplified APIs for the ESG Dataset Catalog, a data lake and metadata platform managing 10M+ artifacts, driving adoption from 1 to 8 teams across New York and London.
- Architected and led an 8-month-long cross-datacenter migration to achieve DR-1 compliance, aligning 6 ESG teams on phased replication and dual writes rather than 6 independent migrations.

SKILLS & TECHNOLOGIES

Languages: Golang, Python, SQL, TypeScript, JavaScript, Lua

Backend & Distributed Systems: PostgreSQL, Kafka, REST APIs, OpenAPI, Chi, sqlc, SQLAlchemy

Infrastructure & Data: Docker, Argo Workflows, AWS S3, IAM, pandas, PyArrow, Parquet

Frontend: React, HTMX, Material UI, Vite, Webpack, HTML, CSS

Testing & Observability: Go test, Pytest, Pandera, OpenTelemetry, Grafana, CI/CD

EDUCATION

SUNY at Buffalo | B.S. Biomedical Sciences

Aug 2015 – Jun 2019

App Academy | Full-Stack Software Engineering Program

Aug 2021 – Nov 2021